

Module 2 - Two-Line Revision Notes (Ollama)

LLM API Fundamentals

LLMs are accessed through stateless APIs. Every request should include all required context.

Chat Completion API

Messages are sent with system, user and assistant roles. The model generates the next assistant message.

Message Roles

System defines behavior, user provides input, assistant stores prior replies. Correct roles improve consistency.

Streaming

Streaming returns tokens progressively. It improves responsiveness for long outputs.

Temperature

Controls randomness in responses. Lower is deterministic; higher is more creative.

Top_p

Limits token selection to the most probable cumulative set. Usually tune either top_p or temperature.

Max Tokens

Caps the maximum generated output length. Prevents overly long responses.

Ollama chat()

Used to interact with local chat models. Accepts model name and message history.

Ollama embed()

Creates embedding vectors from text. Used for semantic search and RAG.

Conversation History

LLMs do not remember previous calls. Applications must resend history.

Context Window

Models can process only a limited number of tokens. Older content must be managed.

Sliding Window

Retain recent messages and discard older ones when limits are reached. Useful for long chats.

Summarization

Compress older conversation into a short summary. Preserves important context while saving tokens.

Configuration

Store configurable values outside code. Keep secrets in environment variables or vaults.

Logging

Capture requests and errors for debugging. Never log secrets or sensitive data.

Retries & Timeouts

Retry transient failures with limits. Always configure request timeouts.

Project Structure

Separate prompts, services, models and configuration. Modular design improves maintainability.

Enterprise Best Practices

Validate inputs, validate outputs and reuse prompt templates. Build layered security around the LLM.

Key Parameters

temperature - creativity; top_p - nucleus sampling; max_tokens - output length; stream - token streaming; model - selected LLM; messages - conversation history; system prompt - defines behavior; seed (if supported) - reproducibility.